

Calculation of Photoionization Differential Cross Sections Using Complex Gauss-Type Orbitals

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Accurate theoretical calculation of photoelectron angular distributions for general molecules is becoming an important tool to image various chemical reactions in real time. We show in this article that not only photoionization total cross sections but also photoelectron angular distributions can be accurately calculated using complex Gauss-type orbital (cGTO) basis functions. Our method can be easily combined with existing quantum chemistry techniques including electron correlation effects, and applied to various molecules. The so-called two-potential formula is applied to represent the transition dipole moment from an initial bound state to a final continuum state in the molecular coordinate frame. The two required continuum functions, the zeroth-order final continuum state and the first-order wave function induced by the photon field, have

been variationally obtained using the complex basis function method with a mixture of appropriate cGTOs and conventional real Gauss-type orbitals (GTOs) to represent the continuum orbitals as well as the remaining bound orbitals. The complex orbital exponents of the cGTOs are optimized by fitting to the outgoing Coulomb functions. The efficiency of the current method is demonstrated through the calculations of the asymmetry parameters and molecular-frame photoelectron angular distributions of H_2^+ and H_2 . In the calculations of H_2 , the static exchange and random phase approximations are employed, and the dependence of the results on the basis functions is discussed. © 2017 Wiley Periodicals, Inc.

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Introduction

Ab initio methods for electronic bound states have been extensively developed and successfully applied to various problems. Conversely, methods for electronic continuum states are still in a comparatively early stage of development. One of the quantities which require electronic continuum states is photoelectron angular distribution. Its measurement has made it possible to observe the motions of valence electron dynamics,^[1] molecular orientations,^[2] and molecular structures.^[3] In fact, recent theoretical calculations^[4–6] show that real time chemical reactions can be tracked by observing photoelectron angular distributions. In the near future, various chemical reactions will be measured in real time by these techniques. To analyze these experimental data, accurate and general calculation schemes for photoelectron angular distributions that include electron correlation effects will be necessary. For this purpose, a calculation method that is easily combined with current quantum chemical techniques is desirable. In this article, we will demonstrate such a direction using complex Gauss-type orbitals (cGTOs).

Photoionization is a half-collision process. Recent rapid developments in experimental fields, in particular using intense pulse lasers have prompted theoretical developments where photoionization is treated as a time-dependent phenomenon in a nonperturbative regime, thus tracking dynamics of electron wave packets. An example of the current theoretical status is described in Ref. [7]. Historically, most half-collision processes including photoionization have been studied based on time-independent scattering theory in the perturbation regime. The direct approach is to solve the

Schrödinger equation to obtain the final continuum state using the stationary state scattering theory. The equation is typically solved with the boundary condition that the wave function becomes a plane wave plus ingoing spherical waves in the asymptotic region. Another approach for half-collision processes is to solve the driven-type Schrödinger equation, which is a modified version of the Schrödinger equation with an inhomogeneous driven term under the outgoing boundary conditions. Band and Freed^[8,9] applied a grid method involving the driven-type Schrödinger equation to photodissociation processes. McCurdy et al.^[10] calculated double photoionization cross sections using an exterior complex scaling method with numerical basis functions. Considering applicability to general molecules, a method based on the so-called L^2 (square integrable) methods is preferable. The complex scaling method^[11,12] and the complex basis function (CBF) method^[13,14] can be interpreted as L^2 methods to solve the driven-type Schrödinger equation. In the CBF method, complex L^2 basis functions such as cGTOs and complex Slater-type orbitals (cSTOs) are used as basis functions.

In the time-independent approach, various techniques^[15–18] have been proposed for treating the boundary conditions of the Schrödinger equation. In the complex Kohn method,^[17] the continuum wave function is calculated by a mixture of

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many L^2 basis functions and a small number of non- L^2 basis functions that satisfy the exact boundary conditions. In the iterative Schwinger variational method,^[15] the boundary conditions are included in the integral equation for continuum wave functions. In these methods, special numerical calculations are necessary. Conversely, in the CBF method, the total cross section can be extracted from diagonalization of the complex symmetric Hamiltonian matrices. There is no need to explicitly include the boundary conditions as for the direct methods; in fact, the procedure for calculating the total cross section in the CBF method is the same as that of the ordinary bound excited state calculations except that the basis functions are complex valued. Therefore, ordinary quantum chemistry algorithms, including electron correlation effects such as CI^[19] and EOM-CC, can be applied to calculations of photoionization total cross sections. Additionally, a well-developed algorithm^[20,21] for evaluating time consuming two-electron integrals can be used if cGTOs are employed. Finally, cSTOs and cGTOs can represent high energy oscillating behaviors better than the corresponding real basis functions, because CBFs themselves have oscillating behaviors with complex orbital exponents. For these reasons, the CBF method can be one of the most powerful methods for computing photoionization total cross sections including electron correlation effects for general molecules.

While total cross sections can be evaluated directly, differential cross sections or photoelectron angular distributions are considered to be difficult to evaluate with the CBF method, since this method does not calculate the continuum orbital explicitly. In fact, many years ago, McCurdy and Rescigno analyzed and developed a method to compute a continuum wave function using the CBF method; they successfully calculated the so-called asymmetry parameter for the H atom photoionization problem, but with many basis functions.^[14] In a recent study, we optimized five or six cSTOs for the frequency dependent polarizability of H atom and found that the optimized cSTOs can accurately represent the solutions of the driven-type equation in the molecular region.^[22] However, this optimization method does not work for cGTO basis functions. In this study, we used an alternative method to represent the solutions of the driven-type equation in terms of several cGTOs by fitting the complex orbital exponents to the outgoing Coulomb functions.

Photoionization differential cross sections can be calculated using the transition dipole moment from an initial bound state to a final continuum state within the dipole approximation. Similar to the work of Band and Freed,^[8] Reinhardt,^[12] and Han and Yarkony,^[23] we used the two-potential formula to evaluate the transition moment. In this formulation, three wave functions are required: an initial bound state wave function, a first-order perturbed wave function, and a set of the zeroth-order continuum state wave functions for the one-center expansion of a final continuum state. The initial bound state wave function can be obtained using ordinary quantum chemistry techniques with a real GTO set, and the first-order perturbed wave function has been obtained routinely as a solution of the driven-type Schrödinger equation using the

CBF method. If the solution for the zeroth-order continuum state can be represented with GTOs, the transition dipole moment can be evaluated using GTOs alone without relying on numerical integrations, thereby realizing a preferable computational method. We relied on the property that the continuum state that satisfies the Schrödinger equation can be calculated from the imaginary part of the solution functions of the driven-type Schrödinger equation. The corresponding driven equation is solved using the CBF method, and the approximate continuum state is obtained in an expanded form with cGTOs basis functions.

As explained before, one of the advantages of the CBF method is that excited state calculation methods for electronic bound states can be easily applied to continuum states. Recently, time-dependent density-functional theory (TDDFT)^[24] has been routinely applied to various bound excited states, and is becoming the method of choice for large molecules. TDDFT has also been applied to continuum states by some groups, albeit to a lesser extent.^[25,26] In this article, we applied the static exchange (STEX) approximation^[27] and the random phase approximation (RPA).^[28–30] The RPA, also called the time-dependent Hartree–Fock method, is based on linear response theory, similar to TDDFT. The total photoionization cross sections of simple molecules were previously calculated with a combination of the RPA and CBF methods; however, no attempt was made to calculate the differential cross sections and anisotropy parameters. In this study, before challenging larger molecules, we used the simpler RPA to investigate the applicability of linear response theory with the proposed method to evaluate differential cross sections and anisotropy parameters.

In the Theory section, we present a method to calculate photoionization differential cross sections and asymmetry parameters using the CBF method. In the Determination of cGTO Basis Function section, we demonstrate a method to formulate efficient cGTOs for solving the driven-type Schrödinger equation using a fitting method. We then show the efficiency and accuracy of the optimized cGTOs by applying them to a simple problem. In the following section, Calculation of Differential Cross Sections, we present the results of the calculations for H_2^+ and H_2 . The Conclusion section summarizes this article.

Theory

Photoionization cross sections

Let us consider a situation in which a linear molecule is in a fixed orientation $\vec{\gamma}$ relative to the laboratory frame coordinate system, and is excited from an initial bound state Φ_i to a final continuum state $\Phi_{\mathbf{k}'\vec{\gamma}}^-$ representing a photoelectron with the wave vector $\mathbf{k}'$ and a residual final cation state. Subsequently, we present some basic equations with one-electron notation for simplicity. The corresponding equations applicable to general many-electron wave functions can be found in Ref. [23]. Then, the appropriate boundary conditions for $\Phi_{\mathbf{k}'\vec{\gamma}}^-$ can be expressed as follows,

$$\lim_{r' \rightarrow \infty} \Phi_{\mathbf{k}'\vec{\gamma}}^{-}(\mathbf{r}') = \sqrt{\frac{k}{(2\pi)^3}} \left\{ e^{i\mathbf{k}' \cdot \mathbf{r}' + i\frac{Z}{k} \log(kr + \mathbf{k}' \cdot \mathbf{r}')} + \frac{f_{\vec{\gamma}}^{-}(\mathbf{k}')}{r} e^{-i(kr + \frac{Z}{k} \log 2kr)} \right\}, \quad (1)$$

where $\mathbf{r}'$ and $\mathbf{k}'$ are the laboratory frame position and wave vectors, respectively, Z is the charge of the residual ion, and $f_{\vec{\gamma}}^{-}(\mathbf{k}')$ is the scattering amplitude. $\Phi_{\mathbf{k}'\vec{\gamma}}^{-}(\mathbf{r}')$ is an energy-normalized continuum wave function. Then, the differential cross sections $d\sigma_{\vec{\gamma}}/d\hat{\mathbf{k}}'$ can be obtained from the transition dipole moment $D_{\mathbf{k}'\vec{\gamma}i}$ as follows,

$$\frac{d\sigma_{\vec{\gamma}}}{d\hat{\mathbf{k}}'} = \frac{4\pi^2\omega}{c} |D_{\mathbf{k}'\vec{\gamma}i}|^2, \quad (2)$$

$$D_{\mathbf{k}'\vec{\gamma}i} = \langle \Phi_{\mathbf{k}'\vec{\gamma}}^{-} | \mu | \Phi_i \rangle, \quad (3)$$

where c is the light speed, ω is the photon energy, and μ is the polarization component of the dipole operator vector. For an ensemble of randomly oriented molecules, the differential cross section is the average of $d\sigma_{\vec{\gamma}}/d\hat{\mathbf{k}}'$ over the orientation $\vec{\gamma}$,

$$\frac{d\sigma}{d\hat{\mathbf{k}}'} = \frac{1}{8\pi^2} \int \frac{d\sigma_{\vec{\gamma}}}{d\hat{\mathbf{k}}'} d\vec{\gamma}. \quad (4)$$

This type of averaged differential cross section is expressed in the following simple expression as first derived by Tully et al.^[31]

$$\frac{d\sigma}{d\hat{\mathbf{k}}'} = \frac{\sigma}{4\pi} [1 + \tilde{\beta} P_2(\cos \theta)] \quad (5)$$

where $\tilde{\beta}$ is the asymmetry parameter, σ is the total photoionization cross section, and $P_2(\cos \theta)$ is the second-order Legendre polynomial in terms of the angle θ between $\mathbf{k}'$ and the polarization vector of the photon field. The angular dependence of photoelectrons averaged over the molecular orientation is fully described by a single parameter $\tilde{\beta}$. It is noted that, as will be discussed later, the differential cross sections for a fixed molecular orientation require terms with higher order Legendre functions. Performing a similar averaging on eq. (4), Tully et al.^[31] presented a compact computational scheme for linear molecules. Herein, we summarize their results for a one-electron system. First, we define a continuum state satisfying the boundary condition represented by the S-matrix $S_{ll'}$. From a computational point of view, this function is conveniently represented by the coordinate $\mathbf{r}$ in the molecular fixed coordinate system and is chosen as an eigenfunction of the z (molecular axis) component of the angular momentum as follows:

$$\Phi_{lm}(\mathbf{r}) \rightarrow k^{-\frac{1}{2}} r^{-1} e^{i\theta_l(r)} Y_{lm}(\hat{\mathbf{r}}) - k^{-\frac{1}{2}} r^{-1} \sum_{l'} S_{ll'}^+ e^{-i\theta_{l'}(r)} Y_{l'm}(\hat{\mathbf{r}}) \quad (6)$$

$$\theta_l(r) = kr - \frac{l\pi}{2} + \frac{Z}{k} \log(2kr) + \eta_l \quad (7)$$

where η_l is the Coulomb phase shift

$$\eta_l = \arg \Gamma \left(l + 1 - \frac{iZ}{k} \right). \quad (8)$$

Since both the molecular frame wave function $\Phi_{lm}(\mathbf{r})$ and the laboratory frame wave function $\Phi_{\mathbf{k}'\vec{\gamma}}^{-}(\mathbf{r}')$ satisfy the same

Schrödinger equation, $\Phi_{\mathbf{k}'\vec{\gamma}}^{-}(\mathbf{r}')$ can be expressed as a linear combination of $\Phi_{lm}(\mathbf{r})$

$$\Phi_{\mathbf{k}'\vec{\gamma}}^{-}(\mathbf{r}') = \sum_{lm} a_{lm}^{\vec{\gamma}}(\mathbf{k}') \Phi_{lm}(\mathbf{r}). \quad (9)$$

Tully et al.^[31] derived an analytic expression for $a_{lm}^{\vec{\gamma}}(\mathbf{k}')$ by comparing the partial wave expansions of $\Phi_{lm}(\mathbf{r})$ and $\Phi_{\mathbf{k}'\vec{\gamma}}^{-}(\mathbf{r}')$ in the asymptotic region. Substituting eq. (9) into eq. (3), followed by averaging eq. (2) with eq. (4), and using angular momentum algebra, the final results are expressed in terms of the molecular-frame symmetry adapted transition moments D_{lm} along with the auxiliary coefficients A_{ξ} for the intermediate angular momentum quantum number ξ as follows,

$$A_{\xi} = \frac{4\pi^2}{c\omega} \sum_{l\lambda m\mu} a_{lm} a_{\lambda\mu}^* (-1)^{-m_l} \frac{\sqrt{(2l+1)(2\lambda+1)}}{4\pi(2\xi+1)} \quad (10)$$

$$\times C(11\xi|-m, \mu) C(11\xi|00) C(\lambda\xi|\mu-m) C(\lambda\xi|00)$$

$$a_{lm} = \omega i \sqrt{\frac{2}{3}} i^{-l} e^{im_l} D_{lm} \quad (11)$$

$$D_{lm} = \langle \Phi_{lm} | r Y_{lm} | \Phi_i \rangle_{m'=m-m_l} \quad (12)$$

where m_i is the magnetic quantum number of the initial state Φ_i and $C(j_1 j_2 J | m_1 m_2)$ are the Clebsch–Gordan coefficients. The total cross sections σ and asymmetry parameter $\tilde{\beta}$ can be calculated from the above coefficients A_{ξ} with $\xi=0$ and 2, as

$$\sigma = 4\pi A_0, \quad (13)$$

$$\tilde{\beta} = \frac{A_2}{A_0}. \quad (14)$$

Han and Yarkony^[23] recently presented a more general formulation.

Driven-type Schrödinger equation

Computing the continuum state $\Phi_{lm}(\mathbf{r})$ is a direct way to solve a photoionization problem; however, there is another way to do this. The first-order wave function $\Psi^{(1)}$ for dipole interaction satisfies the following equation^[32]

$$(E-H)\Psi^{(1)} = \mu\Phi_i, \quad (15)$$

which is called the driven-type (inhomogeneous) Schrödinger equation, because of the additional inhomogeneous term $\mu\Phi_i$. Equation (15) is solved with outgoing boundary conditions for the photoionization problem. Symbolically, $\Psi^{(1)}$ can be written as follows:

$$\Psi^{(1)} = G^+ \mu\Phi_i, \quad (16)$$

where G^+ is the full Green's function

$$G^{\pm} = \lim_{\epsilon \rightarrow +0} (E-H \pm i\epsilon)^{-1}. \quad (17)$$

Here, we briefly discuss the relationship between $\Psi^{(1)}$ and the Schrödinger equation. A more detailed discussion was given in

our previous paper.^[22] Because $\mu\Phi_i$ is a real function, this inhomogeneous term vanishes if the imaginary part is taken for both sides of eq. (15)

$$(E-H)\text{Im}\Psi^{(1)}=0. \quad (18)$$

Thus, the imaginary part of the solution of the driven-type Schrödinger equation satisfies the ordinary Schrödinger equation.

Transition dipole moment

To calculate the transition moment D_{lm} , we applied the so-called two-potential formula. In this article, we developed the formulation for one-electron systems; however, the extension to many-electron systems is straightforward. First, we define the zeroth-order Hamiltonian H_0 and the interaction potential V , which comprise the total electronic Hamiltonian

$$H=H_0+V \quad (19)$$

$$H_0=-\frac{1}{2}\nabla^2-\frac{Z}{r}, \quad (20)$$

where Z is the charge of the residual cation. The partial wave component of the Coulomb function is the eigen function of H_0 , and is given by the following equation:

$$\Phi_{lm}^0(\mathbf{r})=2i k^{-\frac{1}{2}}r^{-1}f_{kl}(r)Y_{lm}(\hat{\mathbf{r}}). \quad (21)$$

Here, $f_{kl}(r)$ is the regular Coulomb function.^[33] Using Φ_{lm}^0 , the continuum state Φ_{lm} satisfying eq. (6) is expressed as follows:

$$\Phi_{lm}=(1+G^-V)\Phi_{lm}^0. \quad (22)$$

As is well-known, the operator $1+G^-V$ is one form of the Møller wave operators. Substituting eq. (22) into eq. (12), we obtain

$$D_{lm}=\langle(1+G^-V)\Phi_{lm}^0|\mu\Phi_i\rangle, \quad (23)$$

which is then converted to the following two-potential formula for half-collision processes,

$$D_{lm}=\langle\Phi_{lm}^0|\mu\Phi_i\rangle+\langle\Phi_{lm}^0|VG^+\mu\Phi_i\rangle=\langle\Phi_{lm}^0|\mu\Phi_i\rangle+\langle\Phi_{lm}^0|V\Psi^{(1)}\rangle, \quad (24)$$

where we use the symbolic expression for $\Psi^{(1)}$ given in eq. (16).

The first term in the final expression in eq. (24) represents a direct contribution to the transition moment that ignores the so-called final state interaction caused by V , whose effect is evaluated in the second term. Note that this final expression is derived using the Lippmann–Schwinger equation, and therefore does not contain any approximation. Reinhardt^[12] and Han and Yarknoy^[23] discussed the use of eq. (23) within the complex scaling method. Band and Freed^[8] solved $\Psi^{(1)}$ using a grid method for photodissociation problems and evaluated differential cross sections using eq. (24).

The total cross section can be simply calculated from the first-order wave function^[11,34],

$$\sigma=-\frac{4\pi\omega}{c}\text{Im}\langle\mu\Phi_i|\Psi^{(1)}\rangle. \quad (25)$$

To calculate the total cross section, only the bound state wave function Φ_i and the first-order wave function $\Psi^{(1)}$ are required, but for the differential cross sections, an additional set of zero-order continuous state wave functions Φ_{lm}^0 is required.

CBF method

There are several methods to treat outgoing boundary conditions for the driven-type Schrödinger equation, such as the complex absorbing potential method,^[35,36] the exterior complex scaling method^[37] and the Sturmian method.^[38] Considering applicability to electronic structure problems, in particular photoionization problems, an L^2 method is preferred. The complex scaling method^[11,12] and the CBF method^[13,14] are such methods. In the CBF method, a real valued electronic Hamiltonian is diagonalized with CBFs

$$(u_n, Hu_m)=\delta_{nm}E_n, \quad (26)$$

$$(u_n, u_m)=\delta_{nm}. \quad (27)$$

Here, the round bracket implies integration without complex conjugation^[13,39] and the eigen energies $\{E_n\}$ are generally complex valued. Using complex energy eigen functions $\{u_n\}$, the wave function $\Psi^{(1)}$ in eq. (15) can be solved variationally as

$$\Psi^{(1)}=\sum_n\frac{u_n(u_n, \mu\Phi_i)}{E-E_n}. \quad (28)$$

As the basis functions for molecular applications, cGTO basis functions are conveniently used. With eqs. (25) and (28), photoionization total cross sections of various molecules have been calculated successfully using a mixture of cGTOs and real GTOs.^[29,40–42]

Determination of Φ_{lm}^0

If Φ_{lm}^0 can be expanded accurately in terms of GTOs, the transition dipole moment D_{lm} in eq. (24) can be exclusively evaluated with GTO basis functions, and any numerical integrations can be avoided in the calculation of differential cross sections and asymmetry parameters. For this purpose, we used the spherical symmetry of H_0 and the property that the imaginary part of the driven equation solution is proportional to the regular solution of the corresponding homogeneous equation.^[43] This feature can be recognized from eq. (18). Because the driven-type equation can be solved by the CBF method, a regular solution can easily be obtained from the imaginary part of eq. (28), as explained below.

The regular Coulomb function $f_{kl}(r)$, which is the radial component of Φ_{lm}^0 as defined in eq. (21), satisfies the following radial Schrödinger equation.

$$(E-h_l)f_{kl}(r)=0, \quad (29)$$

$$h_l(r)=-\frac{1}{2}\frac{d^2}{dr^2}+\frac{l(l+1)}{2r^2}-\frac{Z}{r}. \quad (30)$$

Next, we introduce the corresponding driven equation with an artificial driven term s_l ,

$$(E - h_l)\psi_{kl}^0(r) = s_l(r). \quad (31)$$

We assumed that s_l is real and is localized in the molecular region. The solution $\psi_{kl}^0(r)$ can be represented using the Coulomb Green's function $g_{kl}(r, r')$

$$g_{kl}(r, r') = -\frac{1}{k} f_{kl}(r_{<}) h_{kl}^+(r_{>}), \quad (32)$$

as follows,

$$\psi_{kl}^0(r) = 2 \int_0^\infty g_{kl}(r, r') s_l(r') dr' = -\frac{2}{k} \int_0^\infty f_{kl}(r_{<}) h_{kl}^+(r_{>}) s_l(r') dr'. \quad (33)$$

Here, $r_{<} = \min[r, r']$, $r_{>} = \max[r, r']$, and h_{kl}^+ is the outgoing Coulomb function.^[33] Considering the imaginary part of both sides of eq. (33),

$$\text{Im } \psi_{kl}^0(r) = -\frac{2}{k} \int_0^\infty f_{kl}(r) f_{kl}(r') s_l(r') dr' = -\frac{2}{k} f_{kl}(r) \langle f_{kl} | s_l \rangle. \quad (34)$$

It is obvious that $\text{Im } \psi_{kl}^0(r)$ is proportional to $f_{kl}(r)$. The proportionality constant $\langle f_{kl} | s_l \rangle$ can be determined by taking the inner product with s_l for both sides of eq. (34),

$$\langle s_l | \text{Im } \psi_{kl}^0 \rangle = -\frac{2}{k} |\langle f_{kl} | s_l \rangle|^2. \quad (35)$$

It follows that

$$\langle f_{kl} | s_l \rangle = \sqrt{-\frac{k}{2} \langle s_l | \text{Im } \psi_{kl}^0 \rangle}. \quad (36)$$

Substituting eq. (36) in eq. (34), we obtain

$$f_{kl}(r) = \sqrt{\frac{k}{2}} \frac{-\text{Im } \psi_{kl}^0(r)}{\sqrt{-\langle s_l | \text{Im } \psi_{kl}^0 \rangle}}. \quad (37)$$

From the above expression, the zeroth-order continuum wave function Φ_{lm}^0 can be obtained as

$$\Phi_{lm}^0(\mathbf{r}) = 2^{1/2} i r^{-1} \frac{-\text{Im } \psi_{kl}^0(r)}{\sqrt{-\langle s_l | \text{Im } \psi_{kl}^0 \rangle}} Y_{lm}(\hat{\mathbf{r}}). \quad (38)$$

Now we have expressed Φ_{lm}^0 with the driven equation solution ψ_{kl}^0 , which is practically expressed by the CBF method as shown in eq. (28). We now summarize the previous results for the molecular-frame transition moments as follows:

$$D_{lm} = \frac{i}{\sqrt{2}} \frac{\langle \text{Im } [\psi_{kl}^0] Y_{lm} | r Y_{1m} \Phi_i + V \Psi^{(1)} \rangle_{m'=m-m_i}}{\sqrt{-\langle s_l | \text{Im } \psi_{kl}^0 \rangle}}. \quad (39)$$

To evaluate eq. (39), both ψ_{kl}^0 and $\Psi^{(1)}$ must be calculated accurately. In this article, we obtained these wave functions using cGTOs, thereby their accuracies depend on the L^2 feature. It follows from the expression of eq. (39) that we need an accurate representation of ψ_{kl}^0 in the region where s_l, V and Φ_i are not zero, and an accurate representation of $\Psi^{(1)}$ in the region where V is not zero.

Determination of cGTO Basis Functions

Calculation method

To evaluate eq. (39), efficient cGTOs for the driven-type Schrödinger equation must be set up. For the standard bound state calculations, many GTO sets have been developed for various qualities and purposes. Some authors^[44,45] developed real GTO sets for continuum states that are solutions of ordinary Schrödinger equations. However, the cGTO set for the driven-type equation has not been studied intensively. In a previous study,^[22] we optimized complex orbital exponents of cSTOs for the frequency dependent polarizability of H atom and found that five or six optimized cSTOs can represent the solutions accurately in a typical molecular region. However, this optimization method was not applicable for cGTO basis functions. In another study,^[46] we expanded a cSTO successfully with a linear combination of cGTOs. Therefore, replacing each of the variationally optimized cSTOs with the corresponding cSTO-NG basis function is one way to obtain efficient cGTOs. However, in a typical calculation for a photoelectron with $k=1.0$ a.u., that is, the kinetic energy equal to 0.5 Hartree, the Coulomb function oscillates several times in the molecular region as seen later. Thus, a large number of cGTOs (cGTO-10G, for example) are necessary for a precise representation. In this article, we have pursued an alternative method to set up efficient cGTOs.

As clarified in our previous study,^[22] reflecting its mathematical structure, the solution to the driven equation shows two different behaviors originating from the system independent non- L^2 outgoing wave term and the system dependent L^2 real function term. The latter behavior arises from the inhomogeneous solution term; therefore, it is largely affected by the driven term $\mu \Phi_i$, whose contribution is represented by conventional real GTOs. Therefore, the most crucial step is to efficiently represent the homogeneous outgoing wave term using cGTOs. Accordingly, in this work, a set of complex orbital exponents were optimized by fitting a linear combination of cGTOs to the outgoing Coulomb solution $h_{kl}^+(r)$. For this purpose, we used Prony's method, a signal processing method^[47] used by Huzinaga to develop a STO-NG basis set.^[48] In this method, the distributed radial grid points are predetermined by the increment parameter w and the number of grid points M , as

$$r_m = \sqrt{mw} \quad (m=1, \dots, M). \quad (40)$$

Assuming that the linear combination of N -term cGTOs is fitted exactly to some target function $t'(r)$ at each grid point $\{r_m\}$,

$$\sum_{j=1}^N c_j r_m^{l+1} e^{-\zeta_j r_m^2} = t'(r_m) \quad (m=1, \dots, M). \quad (41)$$

We assume that M is larger than $2N$. If we redefine the target function as $t(r) = r^{-l-1} t'(r)$, the equation becomes

$$\sum_{j=1}^N c_j e^{-\zeta_j r_m^2} = t(r_m) = t_m \quad (m=1, \dots, M). \quad (42)$$

It is convenient to introduce a new set of variables

$$v_j = e^{-\zeta_j w} \quad (j=1, \dots, N) \quad (43)$$

to convert eq. (42) to the following simpler form,

$$\sum_j^N c_j v_j^m = t_m \quad (m=1, \dots, M). \quad (44)$$

There exists an N th-order polynomial whose roots are equal to $\{v_j\}$,

$$\prod_{j=1}^N (v - v_j) = v^N + S_1 v^{N-1} + \dots + S_N. \quad (45)$$

Here, we put $\{S_j, j=1, \dots, N\}$ as the coefficients of the polynomial. From eqs. (44) and (45), for $m=1, 2, \dots, M-N$

$$\begin{aligned} & t_{m+N} + t_{m+N-1} S_1 + t_{m+N-2} S_2 + \dots + t_m S_N \\ &= \sum_{j=1}^N c_j v_j^{m+N} + \sum_{j=1}^N c_j v_j^{m+N-1} S_1 + \dots + \sum_{j=1}^N c_j v_j^m S_N \\ &= \sum_{j=1}^N c_j v_j^m (v_j^N + S_1 v_j^{N-1} + \dots + S_N) = 0. \end{aligned} \quad (46)$$

Equation (46) can be summarized in a matrix form:

$$\begin{pmatrix} t_N & t_{N-1} & t_{N-2} & \dots & t_1 \\ t_{N+1} & t_N & t_{N-1} & \dots & t_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_{M-1} & t_{M-2} & t_{M-3} & \dots & t_{M-N} \end{pmatrix} \begin{pmatrix} S_1 \\ S_2 \\ S_3 \\ \vdots \\ S_N \end{pmatrix} = \begin{pmatrix} -t_{1+N} \\ -t_{2+N} \\ \vdots \\ -t_M \end{pmatrix}. \quad (47)$$

Once the target function values $\{t_m\}$ are given, eq. (47) can be solved numerically using the method of least squares to obtain $\{S_j\}$. From the calculated $\{S_j\}$, $\{v_j\}$ is obtained by obtaining the roots of the polynomial in eq. (45). Finally, the orbital exponents are determined from eq. (43). Using the above method, problems for multiple local minima can be avoided; however, the grid point parameters w and M must be chosen carefully.

Although $h_{kl}^+(r)$ has finite nonzero values at the asymptotic region, cGTOs do not. Therefore, the following window function $w(r)$

$$w(r) = \frac{1}{2} + \frac{1}{\pi} \arctan(-(r-r_0)) \quad (48)$$

was used to make the target function zero at the asymptotic region. Here, r_0 represents the location of the window. We, therefore, determined $\{\zeta_j\}$ with $t(r) = r^{-l-1} w(r) h_{kl}^+(r)$ in eq. (42).

Fitting results

We tried some combinations of the parameters r_0 , w , and N . Figure 1 shows a plot of the (a) real and (b) imaginary parts of the fitted results with five cGTOs, the target function

$w(r) h_{kl}^+(r)$, and the outgoing Coulomb functions $h_{kl}^+(r)$, obtained with $k=1$, $l=1$, $r_0=30$, $w=25$, and $M=100$. Since the first grid point at $r_1 = \sqrt{w}=5$ bohr is so close to the origin, the cGTO fit does not behave as the outgoing Coulomb function $h_{kl}^+(r)$ in the region between 0 and 5 bohr. However, in the region between 5 and 20 bohr, the cGTO fit reproduces the exact function behavior of $h_{kl}^+(r)$. In the outer region, the cGTO fit dies off due to the L^2 feature of cGTOs. By increasing N one by one, we observed that each additional cGTO after $N=5$ resulted in a small amplitude for the cGTO, therefore, we used only five cGTOs.

To analyze the efficiencies of the fitted cGTOs, we solved the radial driven equation

$$\left(\frac{k^2}{2} + \frac{1}{2} \frac{d^2}{dr^2} - \frac{1}{r^2} + \frac{Z}{r} \right) \psi(r) = 2 \exp[-r]. \quad (49)$$

This equation describes the $1s \rightarrow kp$ photoionization of H atom in the velocity form of the transition moment. In addition to the above five cGTOs, fifteen real GTOs were added as basis functions with the orbital exponents distributed from 0.01 to 30 in an even-tempered manner. Figure 2 shows that this GTO set can precisely represent the first-order wavefunction obtained by the grid method up to 15 bohr.

Calculation of Differential Cross Sections

H_2^+

We calculated photoionization cross sections for H_2^+ at the equilibrium internuclear distance of $R=2.0$ bohr. After photoionization, the residual ion has a charge of 2 and we have chosen the zeroth-order Hamiltonian H_0 and interaction V as follows,

$$H_0 = -\frac{1}{2} \nabla^2 - \frac{2}{r}, \quad (50)$$

$$V(r) = \frac{2}{r} + \frac{1}{|r - \mathbf{e}_z R/2|} + \frac{1}{|r + \mathbf{e}_z R/2|}. \quad (51)$$

For the initial bound state wave function, we used a real GTO set of 10s6p4d on each hydrogen atom and the 12s9d set on the center of mass, following a previous calculation reported by Rescigno and McCurdy.^[13] We tried some combinations of the basis sets to expand two continuum functions, $r^{-1} \psi_{kl}^0(r) Y_{lm}(\hat{\mathbf{r}})$ and $\Psi^{(1)}$ in eq. (39), as summarized in Table 1. Using the basis set combinations (a) and (b), we expanded only with the terms $l=1$, $\lambda=1$ in eq. (10), and for (c), (d), and (e), we expanded with terms of $l, \lambda=1$ and 3. In Table 1, 20 p-GTO and 20 f-GTO imply five cGTOs whose orbital exponents are chosen from the fitting calculation to the outgoing Coulomb function at $E=1.0$ a.u. as described in determination of cGTO basis section, and the fifteen real GTOs whose orbital exponents are in a geometric series increasing from 0.01 to 30.0. These large size cGTOs are located on the center of mass of H_2^+ ion. In basis function (e), we further added the two real s-GTOs whose orbital exponents for $\Psi^{(1)}$ are 3.0 and 9.0 on H atoms. As a driven term for the driven equation of ψ_{kl}^0 in eq. (31), we chose a spherical GTO whose exponent is unity

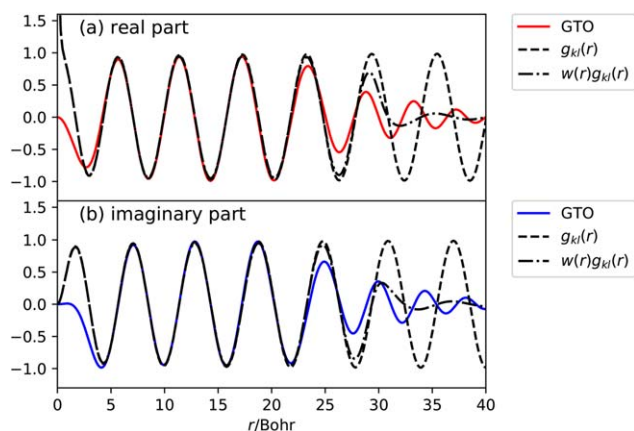


Figure 1. The results of fitting five cGTOs to $w(r)h_{kl}^+(r)$ at $k=1$, $l=1$, $r_0=30$, $w=25$, $N=5$, and $M=100$. Descriptions of these parameters are given in the text. a) Real part and b) imaginary part. The fitting result at $E=0.5$. [Color figure can be viewed at [wileyonlinelibrary.com](#)]

$$s_l(r) = r^{l+1} \exp[-r^2]. \quad (52)$$

We represented the f symmetry cGTOs as linear combinations of Cartesian cGTOs, and evaluated the necessary matrix elements using the McMurchie–Davidson recursion formula.^[21] The necessary incomplete gamma functions for complex arguments were evaluated using the method developed by Ishida.^[20]

Figure 3 shows the results of the $1\sigma_g \rightarrow k\sigma_u$ and $1\sigma_g \rightarrow k\pi_u$ separate channel cross sections and the asymmetry parameters, with the numerical values determined by Brosolo and Decleva^[49] for comparison. Our results were obtained using the length form of the dipole operator, and those with the velocity form differ by about 2% for the cross sections and 1% for the asymmetry parameter at photoelectron energies up to 3.0 a.u.

The middle panel for the $1\sigma_g \rightarrow k\pi_u$ channel shows that all the basis sets have similar accuracies, but detailed comparison suggests that the $l_{\max}=3$ calculations [basis functions (c), (d), and (e)] match the reference values better than the $l_{\max}=1$ calculations [basis functions (a) and (b)].

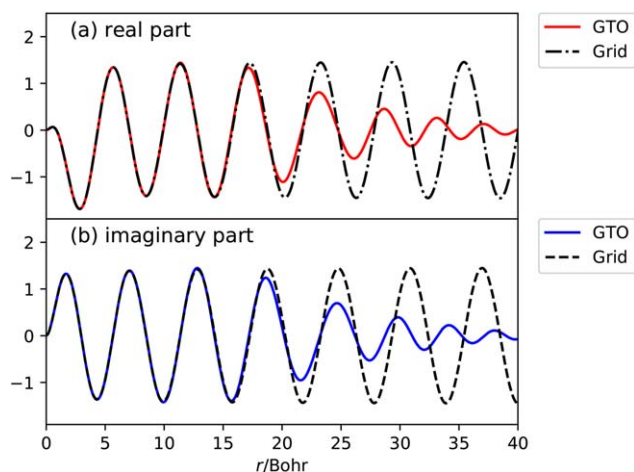


Figure 2. Calculated first-order wave functions of the hydrogen atom photoionization problem by the grid method (Grid) and the CBF method with cGTOs (GTO). [Color figure can be viewed at [wileyonlinelibrary.com](#)]

Table 1. Basis set description to expand $r^{-1}\psi_{kl}^0(r)Y_{lm}(\hat{r})$ and $\Psi^{(1)}(r)$ in eq. (39).					
	$r^{-1}\psi_{kl}^0(r)Y_{lm}(\hat{r})$			$\Psi^{(1)}(r)$	
	$l_{\max}$	$l=1$	$l=3$	Center of mass	On H atom
(a)	1	20 p-GTO		20 p-GTO	
(b)	1	20 p-GTO		20 p-GTO, 20 f-GTO	
(c)	3	20 p-GTO	20 f-GTO	20 p-GTO	
(d)	3	20 p-GTO	20 f-GTO	20 p-GTO, 20f-GTO	
(e)	3	20 p-GTO	20 f-GTO	20 p-GTO, 20f-GTO	2 s-GTO

Contrary to the simple behavior of the $1\sigma_g \rightarrow k\pi_u$ channel cross section (middle panel), we observed a significant basis set dependency in the $1\sigma_g \rightarrow k\sigma_u$ cross section (top panel), as in our previous variational CBF calculations^[50] and also in the asymmetry parameter (bottom panel). The cross sections calculated with $l_{\max}=1$ [basis functions (a) and (b)] have much smaller values than the reference values;^[49] also, the

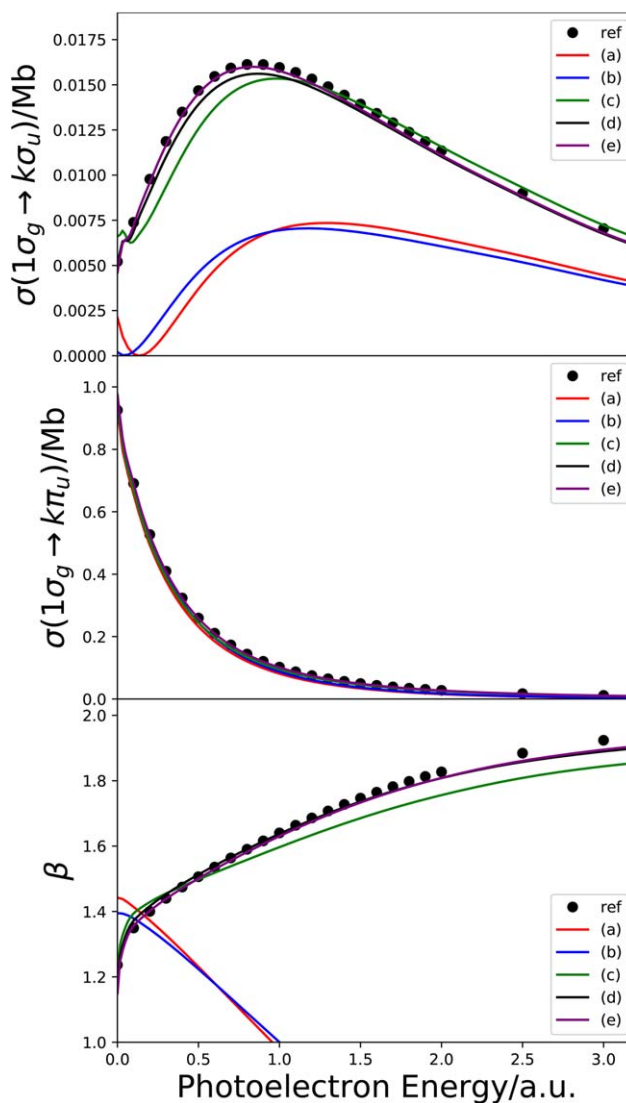


Figure 3. Calculated results for the two separate channel cross sections and the asymmetry parameters for H_2^+ . The plots with solid circles are the results of Brosolo and Decleva.^[49] [Color figure can be viewed at [wileyonlinelibrary.com](#)]

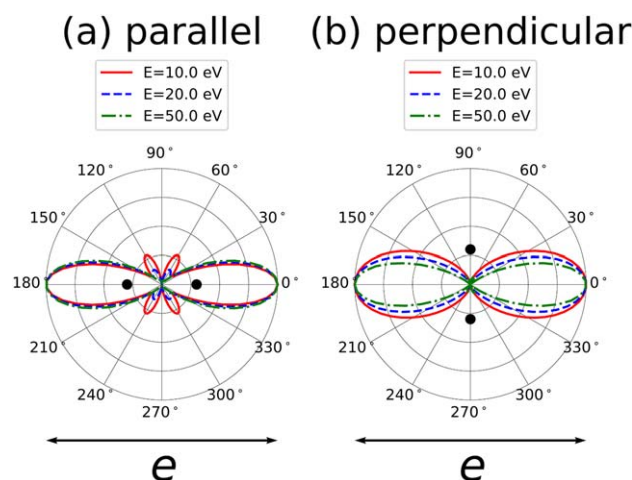


Figure 4. Polar plots of the photoelectron angular distributions of H_2^+ for oscillating electric fields (a) parallel and (b) perpendicular to the molecular axis. The two black dots are the protons, and the vector $\mathbf{e}$ indicates the direction of the oscillating electric field. In these polar plots, the differential cross sections given with respect to the angle measured from the vector $\mathbf{e}$ are normalized so that the maximum value becomes unity at each energy. [Color figure can be viewed at wileyonlinelibrary.com]

asymmetry parameter decreases monotonically with photoelectron energy, which is quite different from that in the reference. The calculations with $l_{\max}=3$ recover qualitative behavior for both the cross sections and the asymmetry parameters [see basis functions (c) and (d)]. These results can be explained by the qualitative difference between the $k\sigma_u$ and $k\pi_u$ continuum wave functions. Because $k\pi_u$ electron has no probability on the protons, the $k\pi_u$ function has a large p-wave component as with the united atom limit. Conversely, the $k\sigma_u$ function has a finite density on the protons, and the photoelectron is scattered and higher angular momentum components become important. In fact, Richards^[51] showed that the $k\sigma_u$ function is dominated by the f-wave component in the low energy region.

The asymmetry parameter obtained using basis function (d) matches the reference values for a wide energy range, but still the $1\sigma_g \rightarrow k\sigma_u$ partial cross sections do not match in the low energy region. One can notice that by increasing the number of partial waves for $\psi_k^0(r)$ and $\Psi^{(1)}$, the $1\sigma_g \rightarrow k\sigma_u$ partial cross sections become more accurate. Although we can add further higher angular momentum basis functions, then we would need complicated linear combinations of Cartesian cGTOs to represent spherical cGTOs. Alternatively, we chose to add the s-GTOs on the H atom in basis function (e) and the calculation provides a nearly converged result. The asymmetry parameters can be obtained accurately with basis functions (c), (d), and (e), with $l_{\max}=3$. Since the $k\sigma_u$ electron is scattered from two protons, the nuclear-electron cusp structure is important to obtain the wave function.^[50] Two s-GTOs added in basis function (e) improve the wave function by expressing nuclear-electron cusp structures. This is one of the advantages of using the multi-center expansion method compared with the single-center expansion method.

Next, we calculated photoionization differential cross sections for an oriented molecule, the so-called molecular-frame

photoelectron angular distributions. Figure 4 shows the polar plots of the differential cross sections for oscillating electric fields given (a) parallel and (b) perpendicular to the molecular axis. Compared with previous theoretical calculation results,^[52] our method gives correct differential cross sections for the oriented molecule. While Figure 4b shows that the angular distribution for the perpendicular electric field is dominated by the p component, that for the parallel electric field in Figure 4a clearly shows a significant f component at low energy. This observation is consistent with the above explanation for the different behaviors of the $1\sigma_g \rightarrow k\sigma_u$ and $1\sigma_g \rightarrow k\pi_u$ separate channel cross sections in Figure 3.

H_2

First, we used the so-called static exchange (STEX) Hamiltonian^[27] to calculate the photoionization cross sections of H_2 .

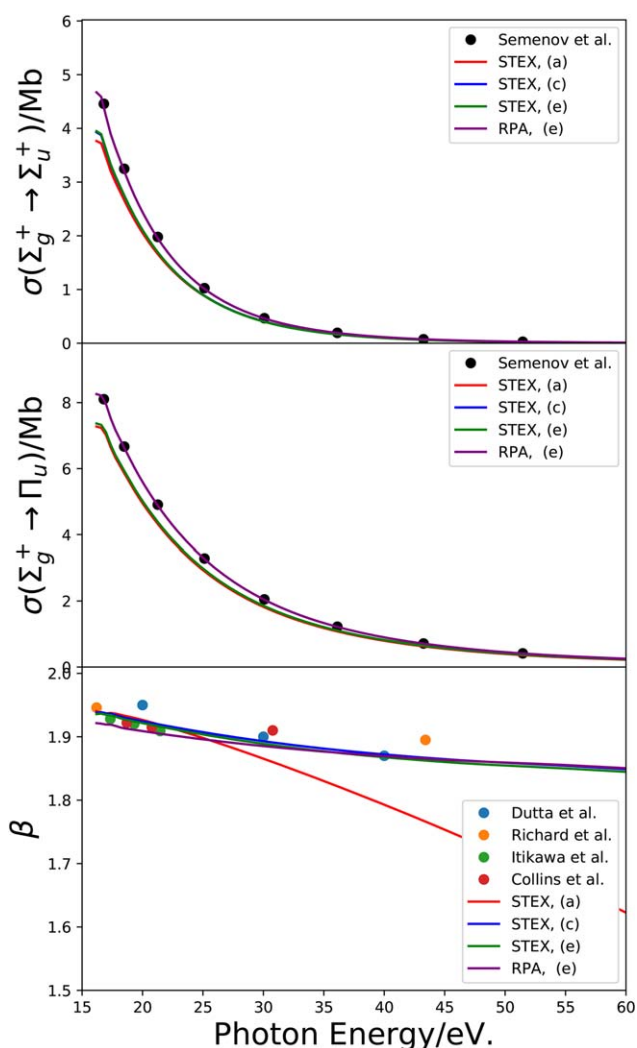


Figure 5. Calculated results for the two separate channel cross sections and the asymmetry parameters of H_2 calculated by STEX and RPA. We used basis functions (a), (b), and (c) for STEX and basis function (e) for RPA. In the upper two panels, the cross sections are compared with those of Semenov et al.^[56] in the bottom panel, the asymmetry parameters are compared with those of Dutta et al.,^[53] Richard et al.,^[51] Itikawa et al.,^[54] and Collins et al.^[55] All results were calculated with the velocity form of the dipole operator. [Color figure can be viewed at wileyonlinelibrary.com]

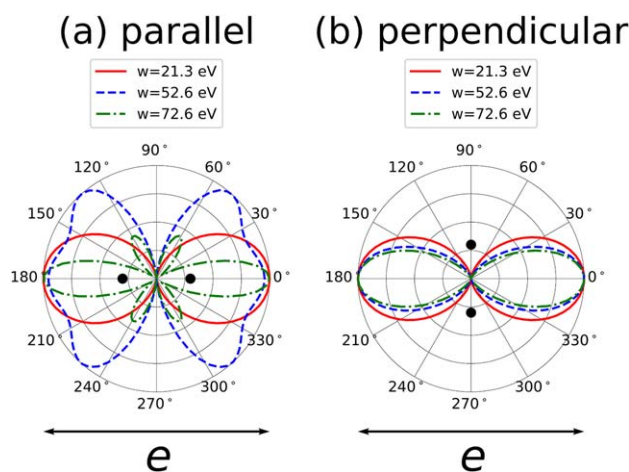


Figure 6. Polar plots of the photoelectron angular distributions of H_2 calculated with the RPA. The same explanation applies as for Figure 4. [Color figure can be viewed at [wileyonlinelibrary.com](http://onlinelibrary.wiley.com)]

Since the charge of the residual ion is unity, we chose the zeroth-order Hamiltonian and interaction potential as follows,

$$H_0 = -\frac{1}{2}\nabla^2 - \frac{1}{r}, \quad (53)$$

$$V(r) = \frac{1}{r} + \frac{1}{|\mathbf{r} - \mathbf{e}_z R/2|} + \frac{1}{|\mathbf{r} + \mathbf{e}_z R/2|} + J_{1\sigma_g} + K_{1\sigma_g} \quad (54)$$

where $J_{1\sigma_g}$ and $K_{1\sigma_g}$ are, respectively, the Coulomb and exchange operators with the Hartree–Fock $1\sigma_g$ orbital for the initial bound state. The uncontracted Huzinaga 10s6p GTOs^[48] on each hydrogen atom were employed as the basis functions for $1\sigma_g$. We calculated the cross sections and asymmetry parameters using the same basis sets described in Table 1.

Figure 5 shows the calculated result obtained with the velocity form of the dipole operator compared to other theoretical results.^[51,53–56] The two separate channel cross sections are converged with respect to the basis functions. Especially, those of the $\Sigma_g^+ \rightarrow \Sigma_u^+$ channel are well described with only the p-wave [basis functions (a) and (b)], although the previous results for H_2^+ required both the p and f waves [basis functions (c), (d), and (e)]. The values of the asymmetry parameter suggest that the error using only the p-wave [STEX basis function (a)] can be largely reduced by employing both p and f GTOs [STEX basis functions (c) and (e)].

The $\Sigma_g^+ \rightarrow \Sigma_u^+$ channel cross sections of H_2 have a quite different shape from that of the corresponding $1\sigma_g \rightarrow k\sigma_u$ channel of H_2^+ . The basis function dependencies of their cross sections also differ significantly. These differences can be explained by the qualitative differences in their $k\sigma_u$ wave functions, that is, while the dominant partial wave for H_2^+ is the f wave, the one for H_2 is the p wave. According to Picca,^[52] this difference arises from the difference in the internuclear distances of two protons. This difference in the $k\sigma_u$ orbital causes significantly different behavior of the asymmetry parameter; H_2 takes a value much closer to 2 compared to H_2^+ . Considering the united atom limit of H_2 , both the $\Sigma_g^+ \rightarrow \Sigma_u^+$ and $\Sigma_g^+ \rightarrow \Pi_u$ channels correlate to the $S \rightarrow P$ channel of the He

photoionization with the asymmetry parameter 2 exactly. Since both $k\sigma_u$ and $k\pi_u$ of H_2 are dominated by the p wave with a small higher angular momentum component, the asymmetry parameter becomes close to 2.

There remains a small discrepancy between the converged STEX cross sections [STEX (a), (c), and (e) in the top and middle panels of Fig. 5] and the data of Semenov et al., who used the random-phase approximation (RPA).^[56] In ordinary bound excited state calculations, linear response theory, such as RPA and TDDFT methods,^[24] is widely applied including to large molecules. To extend the applicability of the present treatment to larger molecules, a calculation scheme based on linear response theory is important. To clarify the source of the discrepancy found in Figure 5, and to ascertain the applicability of the RPA in the evaluation of the differential cross sections and anisotropy parameters, we developed the calculation scheme that is briefly summarized in the Appendix.

Figure 5 also includes the separate channel cross sections and the asymmetry parameters obtained with the RPA using basis function (e) in Table 1. These RPA results are in good agreement with the cross sections of Semenov et al.^[56] and the asymmetry parameters of previous theoretical calculations.^[51,53–55] All the results in Figure 5 were obtained using the velocity form of the dipole operator. Note that the relative difference between the RPA cross sections calculated with the length and velocity forms was only within 1%, while in the STEX calculations, the relative difference was about 30% for the partial cross sections and about 3% for the asymmetry parameters. This gauge insensitivity is one of the important properties satisfied by the RPA.

Finally, molecular fixed photoelectron angular distributions of H_2 were calculated using the RPA. Figure 6 shows the results for oscillating electric fields given (a) parallel and (b) perpendicular to the molecular axis. Our results are in good agreement with the reference results and experimental values.^[56] We emphasize again that our method can be easily applied to general molecules. In particular, Figure 6a shows that at a low photon energy of $\omega = 21.3$ eV, the $k\sigma_u$ orbital is dominated by the p wave. However, as the photon energy increases, the relative weight of the p/f components is switched over, and at $\omega = 72.6$ eV, the $k\sigma_u$ orbital is dominated by the f wave, as was first clarified by Semenov and Cherepkov.^[56] This is the origin of the poor accuracy of the asymmetry parameters using basis function (a) in the high energy region (Fig. 5 STEX (a), bottom panel).

Conclusion

The calculation of differential cross sections of any processes containing continuum states requires that the wave functions satisfy the appropriate boundary conditions. This requirement has long been an obstacle to some L^2 methods. To solve this problem with the CBF method, we propose a calculation method for photoelectron angular distributions with a mixture of complex GTOs and conventionally used real GTOs for representing continuum orbitals and bound state orbitals, respectively. First, the transition dipole moment from an initial bound state to a final continuum state is expressed as a sum of the Coulomb wave term and the correction term based on the two-potential formula. Second, the

zeroth-order continuum state is expressed as the imaginary part of the solution to the driven-type Schrödinger equation. Finally, the transition dipole moment is evaluated using the CBF method.

We set up efficient cGTOs for solving the driven-type Schrödinger equation by fitting the cGTOs to the outgoing Coulomb function. An accurate complex valued solution to the driven-type Schrödinger equation was obtained using the optimized cGTOs.

The method was applied to the H_2^+ photoionization problem and accurate results for the partial cross sections, the asymmetry parameters, and the molecular-frame photoelectron angular distributions were obtained. Applied to the H_2 photoionization problem, the method also provides accurate cross sections and asymmetry parameters, and the molecular-frame photoelectron angular distributions using the RPA.

Acknowledgments

The computations were partly performed using the computer facilities at the Research Center for Computational Science, Okazaki National Institutes.

Appendix

Here, we summarize our calculation scheme using the RPA.^[57,58] Before the specific discussion of the RPA, some general treatments for the appropriate boundary conditions for photoionization processes are explained. As is well-known, the total photoionization cross section from the initial-state wave function with energy E_0 is calculated from the imaginary part of the following matrix element of the full Green's function,^[14]

$$\begin{aligned}\sigma(\omega) &= -\frac{4\pi\omega}{c} \text{Im} \lim_{\eta \rightarrow +0} \langle \Phi_0 | \mu (E_0 + \omega - H + i\eta)^{-1} | \mu | \Phi_0 \rangle \\ &= \frac{4\pi^2\omega}{c} \langle \Phi_0 | \mu \delta(E_0 + \omega - H) | \mu | \Phi_0 \rangle.\end{aligned}\quad (\text{A1})$$

As discussed in the Photodissociation Cross Sections section in the main text, the appropriate boundary condition to resolve the degenerate continuum states with the energy of $E_0 + \omega$ is the incoming boundary condition specified with a superscript $(-)$. Moreover, to discuss the angular distributions of a photoelectron associated with the residual ion state n , the final overall states should be specified with quantum numbers of n , l , and m , as $\Phi_{nlm}^{(-)}(E)$. Since these states satisfy the completeness relation

$$\delta(E - H) = \sum_{nlm} |\Phi_{nlm}^{(-)}(E)\rangle \langle \Phi_{nlm}^{(-)}(E)|, \quad (\text{A2})$$

the total cross section in eq. (A1) is divided into the partial wave components for each channel n ,

$$\begin{aligned}\sigma(\omega) &= \frac{4\pi^2\omega}{c} \sum_{nlm} \langle \Phi_0 | \mu | \Phi_{nlm}^{(-)}(E_0 + \omega) \rangle \cdot \\ &\quad \langle \Phi_{nlm}^{(-)}(E_0 + \omega) | \mu | \Phi_0 \rangle\end{aligned}\quad (\text{A3})$$

The transition moments appearing above play a crucial role in discussing the partial differential cross sections, and are expressed with the two-potential formula,

$$\langle \Phi_{nlm}^{(-)} | \mu | \Phi_0 \rangle = \langle \Phi_{nlm}^0 | \mu | \Phi_0 \rangle + \langle \Phi_{nlm}^0 | V | \Psi^{(1)} \rangle. \quad (\text{A4})$$

Here, the energy is suppressed for simplicity, and Φ_{nlm}^0 is the solution of the zeroth-order Hamiltonian H^0 which consists of a simple sum of the two parts of Hamiltonian, the one given in eq. (30) for the photoelectron and the other for the residual ion, and $V = H - H^0$. $\Psi^{(1)}$ is the first-order wave function $\Psi^{(1)} = G^{(+)} \mu \Phi_0$.

In the RPA formulation, $\Psi^{(1)}$ generated by a time-dependent perturbation of an angular frequency ω is expressed as a linear combination of the one-electron excitation and de-excitation configurations, as follows

$$\Psi^{(1)} = O_\omega^+ \Phi_0, \quad (\text{A5})$$

$$O_\omega^+ = \frac{1}{\sqrt{2}} \sum_{ai} (Y_{ai}^\omega E_{ai} - Z_{ai}^\omega E_{ia}). \quad (\text{A6})$$

Here, Φ_0 is the closed-shell Hartree–Fock wave function for the initial bound state. The indices i and a refer to the occupied and unoccupied spatial orbitals, respectively. The operators E_{pq} are the spin-averaged excitation operators defined by the creation and annihilation operators as $E_{pq} = a_{pz}^\dagger a_{qx} + a_{p\beta}^\dagger a_{q\beta}$; thus E_{ai} and E_{ia} represent the one-electron excitation and de-excitation, respectively. The coefficients Y_{ai}^ω and Z_{ai}^ω are determined by solving the following set of linear equations,^[57,58]

$$\begin{bmatrix} \omega 1 - \mathbf{A} & -\mathbf{B} \\ -\mathbf{B} & -\omega 1 - \mathbf{A} \end{bmatrix} \begin{bmatrix} \mathbf{Y}^\omega \\ \mathbf{Z}^\omega \end{bmatrix} = \begin{bmatrix} \boldsymbol{\mu}^e \\ \boldsymbol{\mu}^d \end{bmatrix}, \quad (\text{A7})$$

where the elements of $\mathbf{A}$, $\mathbf{B}$, $\boldsymbol{\mu}^e$, and $\boldsymbol{\mu}^d$ are given as follows,

$$A_{ai,bj} = (\varepsilon_a - \varepsilon_i) \delta_{ab} \delta_{ij} + 2(ib|aj) - (ib|ja), \quad (\text{A8})$$

$$B_{ai,bj} = 2(ij|ab) - (ij|ba), \quad (\text{A9})$$

$$\mu_{ai}^e = \sqrt{2}(a|\mu|i), \quad (\text{A10})$$

$$\mu_{ai}^d = \sqrt{2}(i|\mu|a). \quad (\text{A11})$$

Here, ε_p is the energy of the canonical Hartree–Fock orbital p , and the two-electron integrals $(pq|rs)$ are in the 1212 (physicists') notation.

Since the zeroth-order final state Φ_{nlm}^0 is the solution of H^0 defined above, it is expressed as an anti-symmetrized product of the imaginary part of ψ_{lm}^0 derived from the driven equation in eq. (31), and the wave function for the residual ion specified by the channel index n . Therefore, in the RPA formulation, Φ_{nlm}^0 takes the following form using only the excitation operators,

$$\Phi_{nlm}^0 = \frac{1}{\sqrt{2}} \sum_a y_{ai} E_{ai} \Phi_0. \quad (\text{A12})$$

The coefficient y_{ai} can be determined from the overlaps between ψ_{lm}^0 and the unoccupied canonical orbitals. The above eq. (A12) relates the channel number n to the occupied orbital i . The channel interactions between two different hole states i and j are accounted for by the off-diagonal matrix elements in eqs. (A8) and (A9) for the $\mathbf{A}$ and $\mathbf{B}$ matrices, respectively. It is

interesting to point out that our final expressions for the partial differential cross sections calculated with the transition moments in eq. (A4) have a similar structure to that of eq. (19) in Ref. [14].

By ignoring the de-excitation contributions $-\omega_1 - A$ and $-B$ in eq. (A7) and by limiting the sum over the occupied orbitals i in (A6) to a particular orbital, the STEx expression is obtained.

Keywords: photoionization cross section • asymmetry parameter • complex basis function method • complex Gauss-type orbital

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